HP / इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED	**Residue up-gradation facility (RUF)** **EPCC-3 package for** **Visakh refinery modernization project (VRMP)**	L&T Hydrocarbon Engineering **L&T- CHIYODA LIMITED**
Title : B016-RUFV-LT-504-QA-QD-0003 **Doc. No.: SHOP QC AND INSPECTION PROCEDURE**		**Rev. No.: 02**

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PMC : **ENGINEERS INDIA LIMITED, NEW DELHI (EIL)**

UNIT : **504**

PMC JOB NO. : **B016**

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SHOP QC AND INSPECTION PROCEDURE

Revision Location Identification Sheet

Sr. No.	Rev. No.	Page No.	Clause No.	Brief Description/ Reason of Revision
1	01	Cover Page	----	Tender Doc. No & LOA No included
2	01	8	5	Revise precedence as per EIL comment
3	01	10,11	8.1	EIL comments incorporated
4	01	12	8.2	EIL comments incorporated
5	01	13	8.6	EIL comments incorporated
6	01	14	9	EIL comments incorporated
7	01	17	12	EIL comments incorporated
8	01	21	14.2,14.3	EIL comments incorporated
9	01	27	18.2.1,18.2.2	EIL comments incorporated
10	01	28	19.3	EIL comments incorporated
11	01	33	23	EIL comments incorporated
12	02	10,11	8.1	EIL comments incorporated
13	02	12	8.2	EIL comments incorporated
14	02	14	9	EIL comments incorporated
15	02	21	14.3	EIL comments incorporated

Contents

1 TERMS and DEFINITIONS 3
1.1 ABBREVIATIONS 5
2 PURPOSE / OBJECTIVE 7
3 SCOPE 7
4 REFERENCE DOCUMENTS 7
5 DOCUMENT PRECEDENCE 7
6 PROCEDURE 8
7 REVIEW CODES 8
8 DOCUMENT APPROVAL & SCOPE OF INSPECTION 9
9 STATUTORY INSPECTION 9
10 List of Documents / Procedures to be submitted by the supplier 14
11 ROLE OF INSPECTOR 14
12 APPROVED TPIA 15
13 ITP/ QAP/QCP 17
14 Acceptance of Structural material and other Miscellaneous items 17
14.1 Steel Structures 20
14.2 Piping 20
14.3 Electrical & Instrumentation 21
14.4 Insulation 21
14.5 Painting 21
14.6 Fireproofing 21
14.7 Civil / Buildings 21
14.8 General 22
15 IDENTIFICATION AND TRACEABILITY 22
16 PROCESS CONTROL 23
17 PRE-INSPECTION MEETING 23
18 INSPECTION NOTICE PERIOD HOLD, WITNESS & REVIEW POINTS OF ITP 24
18.1 CONTRACTOR APPOINTED TPIA 26
18.1.1 Within INDIA 26
18.1.2 Outside INDIA 26
18.2 OWNER/PMC 26
18.2.1 Within INDIA 27
18.2.2 Outside INDIA 27
18.3 INSPECTION LOOK AHEAD SCHEDULE AND CONFIRMATIONS 27
19 MANUFACTURING AND INSPECTION ACTIVITIES AT SUPPLIER'S SHOP 27
20 WITNESS INSPECTION AND TESTS 28
21 SUPPLIER NON-CONFORMANCE 29
22 INSPECTION CLEARANCE 30
23 TPIA DOCUMENTS 31
24 FINAL DOCUMENTATION 32
25 EXHIBIT I – TYPICAL INSPECTION AND TEST PLAN FORMAT 33
26 EXHIBIT II – FLOW CHART FOR NCR MANAGEMENT 34
35

1 TERMS AND DEFINITIONS

Term	Definitions
OWNER	**M/s Hindustan Petroleum Corporation Limited (HPCL), India**
PMC	**M/s Engineers India Limited (EIL).**
CONTRACTOR	**L&T Hydrocarbon Engineering Limited (LTHE).**
LICENSOR	**M/s Chevron Lummus Global (CLG).** **M/s KBR Technology.**
SUPPLIER	The party responsible for the supply of equipment, material or product-related services in accordance with the purchase order issued by CONTRACTOR
ITP/ QCP/QAP	**Inspection and Test Plan / Quality Control Plan / Quality Assurance Plan** Supplier's ITP/QCP/QAP's which includes various inspection activities and scope of Supplier, TPIA, OWNER/PMC's in terms of Review, Witness and Hold points during Inspection. Approval authorities *of ITP/QCP/QAP* will be as mentioned in Table in clause no 8.
PIM	**Pre-Inspection Meetings** A Pre-Inspection Meeting will be held for critical equipment items as identified / agreed in criticality rating procedure (Static/Rotating/Package/Main E&I Equipment) to ensure that technical requirements of the Requisitions are understood by Suppliers, TPIA. The standard PIM agenda shall be followed as per the CONTRACTORs Quality System.
Inspection Category/ Criticality Rating	The level of intervention (QA/QC) to be applied to a CONTRACTOR, Manufacturer or Supplier in order to assure the quality of the procured items taking into account the Criticality rating of their scope of supply.
Inspection Notice	The Project "Inspection Notice" Form will be used by Supplier / Sub Supplier to invite CONTRACTOR / TPIA & OWNER / PMC as per agreed inspection visits.
NCR	Non-Conformance with Project Specification
MDR/ MDB/VDR	**Manufacturing Data Record/ Manufacturing Data Book/ Vendor Data Record** MDR/MDB/VDR's are reviewed by CONTRACTOR/TPIA Inspectors during manufacturing and on completion. MDR include Quality Records in accordance with Project Document, which is attachment to the Requisition.

TPIA	CONTRACTOR appointed Inspection Agency
Hold Point(H)	Hold Point is a stage designated in the ITP/QAP, which requires Witness Inspection before supplier can proceed with further processing, except for cases where prior written permission for further processing or waiver of Witness Inspection by concerned agencies is obtained.
Witness Point(W)	Witness point is a stage designated in the ITP/QAP, which requires Witness Inspection. EPC/LSTK Contractor / Supplier shall perform the activity after proper notification has been given for witnessing the activity. The Contractor/ Supplier is not obliged to hold further processing, unless notified otherwise.
Review Point (R)	Review point is a stage designated in the ITP/QAP which requires the concerned agencies to verify the documents for their correctness and to confirm that the said documents meet the requirements laid down.
Approval Point (A)	A critical step in design, procurement, and or fabrication and testing where it is mandatory that the CONTRACTOR/OWNER/PMC approve a document/qualification/ equipment/or activity. The CONTRACTOR/OWNER/PMC shall be notified at a duration specified in the contract in advance of the first occurrence of the activity or provided the document in a period of time specified by the contract. The activity may not proceed without the CONTRACTOR/OWNER/PMC's approval.
Monitor Point (M)	A critical step in fabrication and testing where it is optional that the CONTRACTOR/OWNER/PMC inspect the component/equipment or witness an activity. The CONTRACTOR/OWNER/PMC shall be notified at a duration specified in the contract in advance of the first occurrence of the activity at each worksite. The activity may proceed with or without the CONTRACTOR/OWNER/PMC's attendance.
IRN/ IC	Inspection Release Note/ Inspection Certificate
REGIONAL PROCUREMENT OFFICE	Inspection office of EIL(PMC),located at various locations.

1.1 ABBREVIATIONS

AI	Authorized Inspector
AS	Alloy Steel
ASME	American Society Of Mechanical Engineers
ASNT	American Society For Non Destructive Testing
BV	Bureau Veritas
CEIL	Certification Engineers International Ltd.
CI	Cast Iron
CRA	Corrosion Resistance Alloy
CS	Carbon Steel
DCI	Document Control Index
DSS	Duplex Stainless steel
E&I	Electrical & Instrumentation
EIL	Engineers India Limited
FAT	Factory Acceptance Test
FIR	Field Inspection Report
GI	Galvanized Iron
HVAC	Heating Ventilation & Air Conditioning
IBR	Indian Boiler Regulations
ICMS	Inspection Call Monitoring System
IR	Inspection Report
IRN/IC	Inspection Release Note/ Inspection Certificate
ISNT	Indian Society For Non Destructive Testing
ITP	Inspection And Test Plan
KOM	Kick Off Meeting
LTHE	L&T Hydrocarbon Engineering Limited
MR	Material Requisition

MRT	Mechanical Run Test
MT	Magnetic Partical Testing
MTC	Material Test Certificate
NCR	Nonconformance Report
NDE	Nondestructive Examination
NDT	Nondestructive testing
PAUT	Phased Array Ultrasonic Technology
PESO	Petroleum and Explosives Safety Organisation
PIM	Pre Inspection Meeting
PMC	Project Management Consultant
PMI	Positive Material Identification
PO	Purchase Order
PQP	Project Quality Plan
PQR	Procedure Qualification Record
PR	Purchase Requisition
PS	Purchase specifiction
PT	Penetration Testing
PVC	Poly Vinyl Chloride
PWHT	Post Weld Heat Treatment
QA/QC	Quality Assurance/Quality control
QAP/QCP	Quality Assurance Plan/ Quality Control Plan
QMS	Quality System Management
RT	Radiography Test
SAT	Site Acceptance Test
SS	Stainless steel
T/T	Tube to tubesheet
TOFD	Time of Flight Diffraction

UT	Ultrasonic Testing
VDR	Vendor Data Record
WBM	Water Bound Macadam
WPQ	Welder Performance Qualification
WPS	Welding Procedure Specification

2 PURPOSE / OBJECTIVE

The objective of this procedure is to:

- Ascertain QA-QC requirements of specification on items manufactured at supplier works or sub supplier works.
- Co-ordination between OWNER/PMC, CONTRACTOR, TPIA, Supplier & Sub-supplier during execution of project.
- General Guideline for - PIM, TPIA Inspector role, Inspection clearance etc.

3 SCOPE

The scope of this procedure is to detail about the procedure and coordination between CONTRACTOR, Supplier / Sub-supplier and OWNER/PMC during Shop Quality Control and Inspection phase for the equipment / items to be supplied for EPCC 03 "RESIDUE UPGRADATION FACILITY UNIT (RUF) Package for VISAKH REFINERY MODERNIZATION PROJECT (VRMP) of M/s Hindustan Petroleum Corporation Limited, Vishakhapatnam (Andhra Pradesh), INDIA.

4 REFERENCE DOCUMENTS

- Project Quality Plan
- EPCC Packages specific Inspection Categorization Procedure
- B016-5-1842-0005 Guideline for Inspection Methodology & Co-ordination Procedure.
- 8-1842-0004 Guideline for Item Categorization Plan

Note: - Latest revision of documents as referred above shall be followed

5 DOCUMENT PRECEDENCE

The order of precedence will be as follows, however in case of conflict - most stringent requirement shall prevail. Supplier shall inform such cases of conflict for LTHE/PMC acceptance / final decision

- Project specific Approved deviation, Agreement, Technical queries (TQs) & Detailed Letter of Acceptance, MoM.
- Item Drawing(s) & Data sheet(s)
- Purchase Order & Purchase requisition
- Inspection and Test Plan (in context of Inspection Interventions).
- Approved QA/QC procedure
- Material Requisition
- Project Specifications and Standards
- Applicable Codes & Standards
- Special Conditions of Contract
- General Terms & Conditions of Works Contract
- Schedule of Prices/ Price Schedule
- Instructions to Bidders
- Other documents.

6 PROCEDURE

CONTRACTOR shall place order(s) on Supplier/Sub Supplier for procurement of Materials and/or Equipments, Packages with the OWNER/PMC Project approved Suppliers List provided by OWNER/PMC and/or as approved by OWNER/PMC in due course of project. OWNER/PMC approval to be obtained for use of Supplier / Sub Supplier which are not appearing in the Approved Supplier List with a specific assessment.

The Materials and/or Equipments, packages are categorized as per approved procedures where CONTRACTOR has assigned an Inspection Category of each items. These items are listed in "Inspection Categorization Procedure" which indicates the Inspection Category. The minimum scope of inspection activities are defined for each inspection Category in clause no. 8. Supplier/Sub-Supplier shall conduct all inspection activities in addition to the contents of this Inspection Categories, as per its own Quality Control Program, Code, Standards & OWNER/PMC Specification requirements.

Supplier shall submit Category of inspection for individual equipment and materials covered under Packages to CONTRACTOR's Inspection Engineer for necessary approvals.

If any Item / Equipment is not listed in Inspection categorization Procedure, supplier shall inform LTHE & obtain necessary category prior to submission of ITP.

7 REVIEW CODES

Approval comments on the submitted documents shall carry one of the following codes:

Code 1	No comments. Proceed with manufacture / fabrication as per the document.
Code 2	Proceed with manufacture/fabrication as per commented document. Revised document required
Code 3	Document does not conform to basic requirements as marked. Resubmit for review.
Code R	Document is retained for Records. Proceed with manufacture / fabrication
Code V	Void - Resubmission required (unless otherwise informed)

8 DOCUMENT APPROVAL & SCOPE OF INSPECTION

Depending on Inspection Category of equipment's and/or materials- QA/QC Documents approval and scope of inspection will be as specified below

All inspection points required by a given class of inspection, shown below, shall be included as a minimum in the ITP which is part of the Manufacturer's /Supplier's overall quality plan.

Supplier shall submit QAP/ITP duly approved by self to CONTRACTOR along with following documents for CONTRACTOR, TPIA and PMCs for approval.

- Un priced Purchase Order copy (Including sub Supplier)
- Approved Purchase Specification
- Approved Drawings / Data sheet in latest revision (as applicable)
- Agreed Deviations, if any.
- Approved Categorization Plan.

8.1 **Inspection Category:** Inspection category of item / equipment's shall be as per the approved Inspection Categorization procedure. Supplier shall ensure inspection category of item /equipment is mentioned in QAQC approval category documents. Supplier shall notify to contractor /PMC if any item / equipment is within the scope of supply but not listed in the approved Inspection Categorization procedure for finalizing the category.

Inspection Categories shall be as per below:

Inspection Category / Class- A:

2 **Scope of Inspection: -** Stage wise and Final Inspection (including document review) witness by TPIA, and witness of *stage* & final inspection, if felt necessary by OWENER/PMC as per the approved ITP/QCP/QAP.

IRN/IC issued by TPIA and to be endorsed by PMC.

Inspection Category / Class - B:

Scope of Inspection: - Final Inspection including document review by TPIA and witness of final inspection if felt necessary by OWENER/PMC as per the approved ITP/QCP/QAP.

IRN/IC issued by TPIA and to be endorsed by PMC.

CONTRACTOR may include additional Critical stage interventions if felt necessary.

Inspection Category / Class - C:

1 **Scope of Inspection: -** *Final Inspection including* document review by TPIA as per approved ITP / QCP /QAP. IRN/IC to be issued by TPIA.

CONTRACTOR may include additional Critical stage and final inspection interventions if felt necessary.

Inspection Category / Class - D:

Scope of Inspection: ITP/QCP/QAP approval not required. OWNER/PMC acceptance is based on CONTRACTOR reviewed MTC / COC / suppliers certificate.

No source inspection required. (Review of test reports and data required by Purchase Order).

For above categories LTHE / PMC / Owner may also participate during inspections along with TPIA.

However for Category A, B, C additional inspection intervention likes Review of supplier QMS, Pre-Inspection meeting, Quality audits etc. will be applied to equipment's / materials as per approved ITP and / or agreed methodology with suppliers.

Activities in respect of QAP / ITP and IRN /IC for categories will be as below:

- For PMC category A & B: QAP/ITP approval by CONTRACTOR, TPIA & PMC. IRN/IC issued by TPIA and to be endorsed by PMC
- For PMC category C : QAP/ ITP approval by CONTRACTOR, TPIA IRN/IC to be issued by TPIA.

1 Documents Approval / IRN Endorsement scope shall be as per below table:

Activities	Contractor	Supplier/ Sub-Supplier	Category A		Category B		Category C	
			TPIA	PMC	TPIA	PMC	TPIA	PMC
ITP	Review	Submit	Approval Required	Approval Required	Approval Required	Approval Required	Approval Required	Approval Not Required
Quality Procedures	Review	Submit	Approval Required	Approval *Not* Required	Approval Required	Approval Not Required	Approval Required	Approval Not Required
Stage Inspection	Not Applicable	Applicable	Applicable	Applicable*	Applicable*	Not Applicable#	Not Applicable#	Not Applicable
Final Inspection	Not Applicable	Applicable	Applicable	Applicable*	Applicable	Applicable*	Applicable	Not Applicable
Issue of IRN/ IC	To endorse TPIA's IRN/IC	To facilitate	To issue	Endorse TPIA's IRN / IC	To issue	Endorse TPIA's IRN / IC	To issue	Not Applicable

* - Inspection as per approved ITP/QCP

- Stage inspection applicable if required as per approved ITP/QC

Note: *The above table is for indigenous items only. For imported items, inspection shall be carried by TPIA only, wherever applicable. Specific OWNER / PMC interventions for overseas inspection shall be informed separately.*

Supplier/ Sub Supplier shall be responsible for obtaining EIL endorsement on IRN issued by TPIA for all category A & B Items / Equipments. Upon PMC endorsements, Original hard copy of IRN shall be submitted to Contractor immediately.

8.2 **PMC endorsement on IRN for below mentioned items is not required:**

- *All Coded U / U2 stamp equipment's. In case Authorized agency scope does not include all the stages as per approved QAP ,the remaining part shall be attended by TPIA*
- All items/ equipment under IBR purview (However for category A & B, wherever partial IBR purview is involved, *Final inspection* & IRN/IC issued by TPIA and to be endorsed by PMC, (*same shall be decided during QAP review*)).
- All imported Items / Equipment.
- All structural & non pressure components of category A & B Items / Equipments.
- Lube oil console/ systems of rotary equipment's.
- Water oil console/ systems of rotary equipment's.
- Loose items & Mandatory & commissioning spares for all category A & B Items / Equipments.

8.3 **For Packaged items: -** Supplier shall submit list of sub-items covered in the package along with category of inspection for CONTRACTOR & OWNER/PMC Approval (as required).

Based on the category rating above, Package Supplier shall submit detailed Categorization plan for the package and relevant Inspection & Test Plans, for sub-ordered items within package, for approval by CONTRACTOR, and OWNER/PMC (as applicable).

8.4 **Pressure Vessels** (if specified in data sheet) shall be ASME U / U2 coded and constructed to in accordance with ASME Section VIII Division 1 or 2 unless specified otherwise in the contract / Purchase Order.

For Coded Equipment's (U/U2): Inspection shall be carried out by AI as well as TPIA. Authorized Inspectors (AI) shall be appointed by Supplier.

8.5 For IBR items:

All statutory requirements as per Indian Boiler Regulations (IBR) shall be met for IBR piping / equipment.

Inspections shall be performed by IBR or by competent person from IBR approved Inspection agency like Lloyds, BV, TUV etc. If supplier intend to utilize services of any IBR approved competent TPIA - valid letter of approval for TPIA and the IBR competent person issued by IBR enabling to carry out inspections on behalf of IBR, shall be provided by Supplier for records and to be furnished (preferably at the time of QAP/ITP approval).

Supplier to submit IBR certificates / Forms in original for supplied equipment's / items. In case of non-availability of original certificate / part supply, photocopy of the original certificate duly attested by the local boiler inspector where the supplier is located is the minimum requirement for acceptance of material.

Well known Pipe manufacturers, as recognized by IBR, shall submit a duly attested copy of Form XVI-G to contractor.

All material shall be supplied along with applicable forms like Form III-A / III-C / II /III/ IV etc. in originals.

All welders used for fabrication of IBR system must possess IBR welding qualification certificate.

QAP for IBR items: Where ever additional TPIA is engaged for IBR items / equipment's, - for clarity of TPIA/PMC scope and interventions, QAP/ITP shall be submitted by supplier to LTHE/EIL- for review/approval. In case of piping bulks relevant notes shall be referred in the standard ITP.

8.6 Electrical Motors

1 ➢ Up to 55KW - No Witness Inspection is required by CONTRACTOR / TPIA / PMC / Owner. *TPIA shall issue IRN based on review of MTC*, Type Test Certificates, Routine Test Certificates and hazardous area certificates.

- 55KW inclusive and above – 100% Routine Test Witness , Type Test Witness (1 per rating / type) and 100% document review including hazardous area certificates by TPIA / PMC / Owner.

9 STATUTORY INSPECTION

Applicable IBR, PESO or any other statutory body's inspection & certification- as applicable shall be provided by Suppliers / sub suppliers, additional inspection by TPIA will be as per approved QAP /ITP. This is applicable for both indigenous and imported items. In case of imported items, inspection will be carried out by Statutory Authority approved TPIA or Statutory authority in the country of the origin.

2 1

- For Motors with VFD operation, Motor shall be certified by statutory for VFD *application* in hazardous application.
- PESO Certificate shall be submitted for combined Motor+VFD tests.
- For Imported Items, ATEX/IECEx Certification for all Ex components shall be in compliance to latest IEC standards and accordingly latest ATEX/ IECEx Certification shall be provided for PESO Certification.
- PESO / IBR and other Statutory Certification shall be under supplier / sub supplier scope

10 LIST OF DOCUMENTS / PROCEDURES TO BE SUBMITTED BY THE SUPPLIER

Following relevant job specific procedures as a minimum shall be submitted (as applicable to the item) by Supplier for necessary review and approval.

- Inspection and Test Plan /QAP /QCP
- NDT Procedures
  - TOFD/ PAUT/UT/RT/MT/PT (As applicable)
  - NDT map with details of % of NDT
- Welding Documents
  - WPS/PQR/WPQ/Weld Map
  - Weld Repair Procedure

- ♦ Consumable and Handling Procedure

➢ Other QA-QC Procedures

- ♦ Hydro Test Procedure
- ♦ Pneumatic test Procedure
- ♦ PWHT Procedure with sketch
- ♦ Ferrite measurement Procedure
- ♦ PMI Procedure
- ♦ Hardness test Procedure
- ♦ Trial Assembly Procedure
- ♦ Pickling and Passivation Procedure
- ♦ Product Test Coupon Procedure
- ♦ T/T Joint welding/Expansion Procedure
- ♦ T/T Joint Mock Up Procedure
- ♦ FAT/SAT/Run Test Procedure
- ♦ MRT/PT/Complete unit Procedure
- ♦ Galvanizing Procedure
- ♦ Surface Preparation and Painting Procedure
- ♦ Preservation and Storage Procedure
- ♦ Refractory Application and Dry-Out Procedure.

NOTE: Supplier shall submit VDR / DCI for Quality document for LTHE approval before start of submission of documents. All documents shall be submitted through portal with proper project specific coversheet and document number.

11 ROLE OF INSPECTOR

(CONTRACTOR / Supplier appointed TPIA)

Inspection shall be carried-out by CONTRACTOR /Supplier appointed TPIA. CONTRACTOR / Supplier shall ensure that the inspectors deployed for inspection are suitable for the scope of work & are competent.

TPIA Inspector will carry out Surveillance, Review, monitoring, Witnesses & Hold Point Inspections as mentioned below as a minimum:

- Conform the Supplier processes, procedures are effective in meeting product requirements, where specified.
- Carry out surveillance, inspection and witnessing sufficient to ensure specified requirements are met and to enable Certification to be issued, where specified.
- Ensure the Supplier personnel including QA/QC personnel are suitably qualified and competent, where specified.
- Ensure proper test & inspection equipment is used, calibrated and certified.
- Ensure all NCRs closed out.
- Review and endorse inspection certificates.
- Review and endorse material test certificates.

Supplier / sub-supplier shall perform and comply to all the inspection requirements for the scope of supply specified in MR/PR/PO.

For inspections of stage / final inspections - LTHE/ OWNER/PMC may also participate during inspection at Supplier's / Sub-Supplier's along with TPIA, The activities of inspection shall be as per approved QAP/ITP/QCP as a minimum. Following additional activities may also be performed:

- Verify the product conformance to contractual requirements at various stages of the fabrication/manufacturing as stated in approved QCP / ITP and applicable technical specifications.
- Review quality documentation to verify product conformance to contractual requirements.
- Carry out surveillance inspections to verify processes and procedures compliance.

The surveillance role shall be as follows but not limited to:

- To monitor manufacturer's/CONTRACTOR's activities with respect to quality performance.
- To ensure that the materials are from the approved suppliers.
- Ensure the Supplier personnel including QA/QC personnel are suitably qualified and competent, where specified.
- To ensure close out of NCRs with suitable corrective and preventive actions.

The acceptance of the CONTRACTOR / supplier appointed TPIA/OWNER/PMC's Inspection personnel (QA/QC Inspector/TPI) shall not absolve the supplier's responsibilities to supply the product that meets contractual requirements.

Note: - Supplier / Sub-Supplier shall obtain signature of OWNER/PMC's inspector and TPIA on QA/QC documents and ITP after completion of inspection activities.

12 APPROVED TPIA

When the inspection is carried out by CONTRACTOR / supplier appointed TPIA, CONTRACTOR / supplier shall use a TPIA that is approved by the OWNER/PMC. CONTRACTOR / supplier shall ensure that the TPI agency is suitable for the scope of work and personnel deployed to certify are competent.

Following is the list of CONTRACTOR / Owner approved TPIA for performing inspection activities at Sub-Supplier / Suppliers works.

- LLOYDS REGISTER
- BUREAU VERITAS
- CEIL
- *TUV* (1)
- DNV

Witnessed Inspection / Review / Inspection waiver by TPIA does not relieve supplier / manufacturer / sub-suppliers from contractual obligations & its responsibilities to supply the item(s) concerned in full compliance with the requirements of the PO, project specification, applicable codes & standards & approved Quality Plan.

13 ITP/ QAP/QCP

PMC/CONTRACTOR's sample Inspection and Test Plans for major items are attached in MR. Post order, Detailed Inspection and test plan shall be prepared by supplier taking into consideration Sample ITPs, Codes, Standards, Material Specifications and Material/Purchase Requisition, OWNER/PMC Specifications, Statutory requirements (if any), lessons learnt. Supplier / Sub- Supplier shall use the format of ITP/QAP/QCP as enclosed in EXHIBIT-I. ITP /QAP/ QCP format of Supplier may be acceptable if all the contents of EXHIBIT-I are covered and is mutually agreed by LTHE and Supplier.

The ITP/QAP/QCP shall give complete listing of stages of inspection, final inspection, tests and material certification requirements for the item, meeting all requirements specified in

the Contract Document / Approved PR / Approved Drawings & Data sheets and shall clearly indicate Hold (H), Witness (W) Inspection, Review (R) stages of inspection to be carried out by TPIA, PMC and OWNER.

For the supplier, all the stages of inspection shall be HOLD points and no stage shall be offered to TPIA / PMC/ OWNER without having performed/witnessed such stages.

The ITP/ QAP/QCP thus generated shall identify and include the following as minimum:

- Sequence of all inspection and test activities related to the item manufactured, specific procedures, methods and work instructions to be applied to each activity in manufacturing of the item.
- This will include but not limited to material identification, qualification of welding, welders, welding operators, NDT procedures & personnel, Post Weld Heat Treatment procedures, Positive Material Identification (PMI), visual / dimensional checks, pressure tests, performance tests, string tests, strip tests, respective acceptance criteria and pre-shipment clearing procedures etc. as the case may be.
- Incoming, in-process and final inspection requirements & it's extent for various agencies.
- Identification of product from applicable drawings, specifications or other documents during all stages of production, delivery and installation and to ensure that each product which is going in to the process of manufacture has proper identification / traceability throughout the process including final output.
- Inspection and test records maintained for each stage of manufacturing.
- All inspection stages such as Hold (H), Witness (W) and Review (R) to be followed and attended by responsible agencies according to approved inspection category of item shall be marked.

CONTRACTOR / TPIA shall ensure that all personnel engaged in performing the activities have requisite qualification and relevant work experience wherever required. Specialized operators shall be qualified and certified.

NDT operators employed, if any, shall be certified as per ASNT/ISNT and shall have valid certificates.

The ITP/QAP/QCP shall be reviewed and signed by Sub-Supplier, Supplier before submitting to CONTRACTOR & *TPIA* for review & approval. CONTRACTOR will obtain PMC approval on this ITP/QAP/QCP as per project requirement.

It is Supplier's prime responsibility to update and maintain ITP/QAP/QCP and other QA/QC procedures as a live document and shall revise and re-submit for approval of CONTRACTOR and OWNER/PMC as and when significant changes are identified in Drawing, MR, Data Sheet etc. and subsequent approval shall be obtained from CONTRACTOR & OWNER/PMC.

In case of multiple ITP/QCP/QAPs for package items, Supplier/Sup-supplier need to provide index - listing out all ITP/QCP details with document number, title and revisions.

Obtaining approval of ITP/QAP/QCP as a minimum in Code-2 by respective agencies is mandatory for the supplier prior to the commencement of any manufacturing activities.

Unless agreed by CONTRACTOR/ OWNER/PMC, supplier shall submit all its major Sub-Supplier's ITP/QCP & other Procedures for approval of CONTRACTOR/ TPIA / OWNER/PMC as minimum. Supplier / Sub Supplier shall follow the approved ITP/ QAP/ QCP.

The CONTRACTOR/OWNER/PMC reserves the right to inspect products or services purchased by the Suppliers from Sub-suppliers. Such inspection shall take place at the source, in the Supplier's premises, on site or at the Sub-supplier's premises.

Verification or Inspection by the CONTRACTOR/OWNER/PMC is not a substitute for verification or Inspection by the Supplier nor absolve the Supplier of its contractual obligations. Hold, Review and Witness Points shall be identified in the Inspection and Test Plans/QAP/QCP for Suppliers, Sub-suppliers, Third Party Inspector / Authorized inspector and OWNER/PMC. Where the OWNER/PMC employs a Third Party Inspection Agency, they shall have the same access to the Supplier's Quality System documentation and reports as the OWNER/PMC.

All materials shall be inspected on receipt by the Supplier prior to TPIA /OWNER/PMC inspection as per ITP scope. Where ever TPIA /Client Hold / witness point for materials, no materials shall be used / installed without CONTRACTOR / OWNER / PMC / TPIA's inspection / written acceptance.

Degraded / deteriorated / earlier used materials shall not be offered for inspection.

For bulks like Piping, Electrical and Instrumentation materials, item specific Standard ITP/QCP shall be prepared by CONTRACTOR considering Project requirements. The standard ITP/QCP shall be approved by OWNER/PMC. For exceptional cases (like exotic materials / nonstandard items etc.) suppliers ITP in line with project requirement shall be reviewed and approved by TPIA / Contractor / PMC.

The Approved ITP/QAP/QCP shall be sent to supplier while pre-order / post order stage for Piping Bulk items. Supplier/Sub-supplier shall follow the approved ITP/QAP/QCP. Suppliers may obtain clarifications, however no comment / deviations will be entertained on the approved document.

In case of suppliers ITP - Supplier's signed & endorsed ITP/QAP/QACs shall be forwarded to CONTRACTOR well in advance before start of manufacturing activities.

Inspection/surveillance of work by TPIA/ CONTRACTOR /OWNER/PMC is intended to verify the effectiveness of the Supplier's own inspection or test activities and such supplementary inspection shall not relieve the Suppliers of their obligations with regard to conformity of the work to OWNER/PMC Specifications, Standards & Code requirements. The Category of CONTRACTOR's inspection/surveillance may change upwards or downwards during the course of the work, depending on Supplier's performances.

14 ACCEPTANCE OF STRUCTURAL MATERIAL AND OTHER MISCELLANEOUS ITEMS

For following items CONTRACTOR will accept these materials based on the review of manufacturer's TC / COC / Guarantee Certificates and physical verification of these materials at site / shop after receipt (including necessary additional tests as specified in the contract / PO / MR). CONTRACTOR reviewed certificates shall be submitted for OWNER/PMC review and acceptance.

14.1 Steel Structures

- Misc. steelworks materials (incl. platforms, etc.) excluding pre-fabricated Structural Steel
- Structural steel material – (structures) Plates, Sections / shapes

14.2 Piping

- Standard pipe support materials (Includes normal CS and supports made from steel section, plate, bar and angle)
- *Field joints* 1
- Valve Positioners
- CI / GI / PVC Pipes, Conduits, GI pipe sleeves
- Hand railing material

14.3 Electrical & Instrumentation

- Mounting posts & misc. steel support materials and brackets
- E&I accessories/ consumables (e.g. bolts, insulation tapes, ferrules, cable lugs, tie wraps, markers, tags, sealing putty, etc.)
- Underground cable identification (colored concrete or cable tiles)
- Limit Switch
- Flexible cables (1.0 – 4.0 sq. mm) – PVC wire *(bare wires with insulation only)* 1 2
- Earthing material
- Hardware used at Site for E&I items
- Lugs & Cable termination kits

14.4 Insulation

- Insulation material (Incl. metal sheeting, vapor barrier, mastic & accessories)
- Insulation including Under / Over Deck material

14.5 Painting

- Touch-up materials of shop applied galvanizing/coating
- Materials for site blasting and surface preparation
- Materials (all coats/systems) for painting structural steel materials, piping materials, cable trays etc.
- Paint material for identification of equipment and piping
- Paints, Chemicals, Catalyst and Cleaning agents
- Oil lubricants
- Wrapping & Coating material

14.6 Fireproofing

- Fire proofing products & accessories (for structural steel, equipment skirts/saddles, cable trays)

14.7 Civil / Buildings

- Concrete, Cement, Grouting Material, Bricks, Fine Aggregate, Coarse Aggregate & all associated civil works materials
- Concrete Admixtures
- Reinforcement bars
- Hume Pipes, Foundation Bolts & Nuts with Sleeves
- WBM materials, Sealant (Bitumen, Polysulphides)
- Sanitary items
- Kitchen equipment
- Ancillary components around buildings (e.g. chemical filtration unit, acid neutralization tank, etc.)
- Furniture
- Buildings fixtures (e.g. doors, glass, glazing, enclosure fencing etc.)
- Ceiling and Exhaust Fan
- Buildings finishes/specialties (e.g. toilets, lockers, partitions, flooring, ceiling, etc.)
- Painting
- Built-in items (e.g. fume cupboards, workbenches, etc.)
- Building Services/utility equipment & materials, incl. civil/plumbing, electrical utility, mechanical.
- Lightning protection materials / grounding for buildings/services
- Bulk Materials (e.g. cables, conduits, cable trays, piping, fittings, supports, etc.)
- Structural steel materials and steel supports
- G.I. sheets for duct work etc.
- HVAC Ducted Zones Reheat Coils (Electrical type)
- HVAC Air Distribution Outlet & inlets
- Foundation Bolts & Nuts with Sleeves

14.8 General

- Fire Extinguishers, Industrial Junction Box, Telephone Junction box , Insulated wires
- Shock Treatment Charts & First Aid Box
- Rubber Mat & Rubber hand Gloves
- Loose spares
- Catalogue items

For items like Structural steel, piping etc. fabricated at site/Pre fabrication Yard- inspection shall be carried out by LTHE's representative / personnel & OWNER/PMC (as applicable).

15 IDENTIFICATION AND TRACEABILITY

Identification and traceability of materials, component and equipment's shall be maintained and documented throughout the pre-fabrication and installation process for all pressure retaining components and non-pressure retaining components welded to pressure retaining components including welding consumables, paints, coatings, adhesives and CRA internals etc. Additional precautions shall be applied for manufacturing AS / SS / DSS / CRA and low temperature materials. For identification stamping, low stress stamps shall be used. Use Chloride free markers in case of marking Stainless Steels. Wherever stamping is impractical / prohibited on components as per project requirement, supplier to propose suitable measures for identification for CONTRACTOR/ PMC acceptance & approval.

The Supplier is responsible for ensuring identification, color coding and material traceability including positive material identification (PMI) meeting the requirements:

Positive Material Identification (PMI), shall be performed as per approved Positive Material Identification procedure.

16 PROCESS CONTROL

All welding and related activities (e. g. Post Weld Heat Treatment and Non Destructive Examination) shall satisfy the requirements of relevant international standards as specified in the contract / purchase order & project specification requirement.

Special processes (e.g. Welding, Painting, PWHT, NDE, Refractory, Concreting, Grouting, Coating, Cable jointing etc.) shall be carried out under supervision of a competent person as defined by the respective specifications. All special processes shall be executed in accordance with documented and qualified procedures that ensure work is complying with OWNER/PMC technical specifications.

The Supplier shall maintain an index or register of all approved welders and NDE operator continuity records which shall be made available to LTHE/TPIA/OWNER/PMC when

requested. The register shall give information on training of the welders and date and results of qualification tests. WPS number, PQP number, base metal, type of welding consumable, joint design, welding position etc. shall also be recorded in the register.

The procedure qualification tests for special processes and the operator (e. g. welder, NDT technician, painter etc.) performance qualification tests shall be witnessed by TPIA.

Process control documents as required by the ITP/QAP/QCP shall be submitted well in advance prior to commencement of the activities for review and approval (as applicable and as stated in the ITP/QAP/QCP). They shall be available at the location where the activities are carried out.

17 PRE-INSPECTION MEETING

The pre-inspection meeting shall be initiated by supplier / Sub-Supplier for Category A,

and as applicable for cat B & C. Equipments with notification to CONTRACTOR & OWNER/PMC through Inspection Call Monitoring system (ICMS).

The Pre-Inspection Meeting shall be conducted at supplier/sub-supplier place. Supplier and his major Sub-Supplier, CONTRACTOR's representative including Quality personnel, TPIA, shall participate in PIM. OWNER/PMC representative may also participate in PIM. Supplier shall provide tentative plan of Four (4) weeks / 30 days for participation in PIM/KOM. Firm notification shall issue in ICMS 7 working days prior to the meeting.

PIM between Supplier / TPIA and LTHE can be conducted by LTHE representative along with LTHE's TPIA representative by visiting supplier works / Tele Conference/ Video Conference etc. In case of Tele conference / Video Conference is arranged for PIM. LTHE to ensure that LTHE appointed TPIA representative is present during PIM. PIM may also be conducted for specific equipment's / items of categories other than 'A' / 'B' which will be informed by LTHE to suppliers.

The pre-inspection meeting should be a part of the pre-fabrication meeting. The basic purpose of pre-inspection meeting is to discuss the supplier scope, purchase order quality requirements in detail, and to verify whether all concerned personnel are familiar with the

requirements and extent of inspection specified in purchase order, relevant specifications, standards, codes and drawings. In addition, witness points and hold points shall be discussed.

Note: Pre-inspection meeting will be excluded for items which are ordered on OWNER/PMC mandatory suppliers, stockiest and for E&I and Piping bulk items. E&I and piping bulk shall also be procured from approved suppliers, as applicable.

Supplier/Sub-supplier shall ensure that Engineering & QA/QC documents necessary for PIM are submitted to CONTRACTOR minimum 30 days advance of schedule PIM date for OWNER/PMC approval.

During PIM following points will be discussed and agreed between SUPPLIER, TPIA, CONTRACTOR & OWNER/PMC

- The scope of shop inspection and tests, availability of applicable project documents, such as purchase orders, requisitions, specifications and Supplier's drawings and procedures.
- Supplier's shop organization QA/QC organogram for the project will be explained by Supplier
- Coordination procedures, such as communication channels among the parties concerned and submission of a Weekly / monthly inspection schedule, applications, records, etc. will be discussed and confirmed at the meeting.
- Communication protocol for QA/QC documents submission / approval.
- Supplier's quality control system to be demonstrated. CONTRACTOR will consolidate the information furnished during the pre-award stage.
- Status of applicable documents viz. Specification, QAP, WPS / PQR, Inspection & Test procedures
- Contract requirements to ensure SUPPLIER's and TPIAs understanding of its responsibilities
- Sequence of manufacturing, testing will be explained by Supplier.
- Inspection and Test Plan / related procedures will be reviewed and finally approved by the OWNER/PMC / the Purchaser.

- Inspection items to be witnessed by the CONTRACTOR and OWNER/PMC and also other matters not confirmed yet, such as hold points and detailed inspection measures, will be taken up and decided.
- OWNER/PMC special requirements if any
- NDE Requirements, Hold points & Witness points
- Inspection notification methodology
- Discussion for Sub- SUPPLIER requirement
- Manufacturing / Inspection Schedule
- SUPPLIER quality control program check; inspection hold points

18 INSPECTION NOTICE PERIOD HOLD, WITNESS & REVIEW POINTS OF ITP

Supplier / Sub Supplier shall forward inspection notification to CONTRACTOR/PMC & TPIA co-coordinator through ICMS prior to the scheduled inspection for CONTRACTOR/TPIA/OWNER/PMC appointed representative for participation in inspection and/or test as per approved ITP/QCP/QAP.

Supplier / Sub-supplier shall fill up all the applicable details clearly specifying the detailed activities etc. in ICMS before sending the notification to CONTRACTOR / PMC & TPIA.

The minimum Inspection notification period for inspection activities like, Review, Witness, Hold, Surveillance etc., by CONTRACTOR/TPIA, OWNER/PMC shall be as given below

18.1 CONTRACTOR APPOINTED TPIA

18.1.1 Within INDIA

- 3 working days advance written notification
- 1 working days advance written notification for inspection re-schedule / cancellation

18.1.2 Outside INDIA

- 30 days tentative advance written notification to LTHE for making travel arrangements.
- 7 working days advance written notification for LTHE / LTHE TPIA

- 4 working days advance written notification for inspection re-schedule / cancellation

18.2 OWNER/PMC

18.2.1 Within INDIA

1

- *10* working days advance written notification for OWNER/ PMC inspection or Test at supplier premises
- 2 (Two) working days advance written notification for any inspection schedule cancellation / rescheduling.
- OWNER/PMC shall provide confirmation for their participation / postponement - 2 working days advance w.r.t. schedule date

18.2.2 Outside INDIA

1

- For any planned inspection or Test at supplier premises (Outside INDIA) - to make travel arrangements, suppliers shall provide min *45* days advance written notification for LTHE/ OWNER/ PMC. Firm / confirmed date shall be notified 10 working days in advance.
- 7 days advance written notification for any inspection schedule cancellation
- OWNER/PMC shall provide confirmation for their participation / postponement - 2 working days advance w.r.t. schedule date

18.3 INSPECTION LOOK AHEAD SCHEDULE AND CONFIRMATIONS

- Supplier shall provide one month look-ahead inspection schedule to CONTRACTOR's Inspection Engineer for TPIA and OWNER/PMC participations. Look ahead schedule to be submitted at least seven working days in advance or as mutually agreed.

Any inspection activities which call for OWNER/PMC inspection as per approved ITP / QCP, Supplier / sub-supplier will clearly mention OWNER/PMC intervention in the Inspection notification format and forward to CONTRACTOR's Discipline Inspection Engineer through Inspection Call Monitoring system (ICMS).

Supplier / Sub-Supplier shall plan the inspection visits required by CONTRACTOR / TPIA in such a manner so as to achieve maximum inspection stages attended with minimum possible inspection visits.

19 MANUFACTURING AND INSPECTION ACTIVITIES AT SUPPLIER'S SHOP

Manufacturing and stage wise inspection may proceed based on CONTRACTOR and OWNER/PMC approved (as applicable) Code-2 (as a minimum) documents like Purchase requisition, Drawings, Data sheets, ITP/QCP and applicable Procedures. However final inspection shall be carried out only with CONTRACTOR & OWNER/PMC approved Code-1 documents only.

1 It is the responsibility of the Supplier / *CONTRACTOR* to furnish copies of the latest revision of the approved specifications, drawings, documents, ITP, approved deviation (if any) & Procedures to TPIA /OWNER/PMC and ensure that the inspection is carried out as per the latest revisions and approved documents. Also, supplier shall be responsible for ensuring availability of all tools and equipment (duly calibrated) that may be required for satisfaction of an inspection / test.

Supplier shall submit all applicable QA/QC and NDT procedures to CONTRACTOR/TPIA & OWNER/PMC for review or approval as identified in the approved ITP / QAP. Procedure may require CONTRACTOR / TPIA / PMC approval for Advanced NDT methods, automatic NDT techniques, cladding applications and any other technique / process. Supplier shall submit the procedures well in advance for necessary approvals.

Welding documents like WPS, PQR those are already approved by any of the inspection agencies listed in clause no. 12 of this document may be used by the Supplier - provided it meets the requirements of this project and shall be reviewed and accepted by the appointed TPIA, CONTRACTOR and / or OWNER/PMC (as applicable).

Similarly welder qualification certificates as per the applicable code requirements shall be acceptable & shall be reviewed by attending Inspector at the time of inspection. For items / equipment's those involve multiple weld joints and welding procedures, supplier/sub-supplier shall prepare detailed Weld Map/Plan and submit it along with WPS and PQR for approval of TPIA, CONTRACTOR, and PMC (as applicable).

Piping materials such as pipes, fittings, flanges, fasteners, Gaskets etc., shall be color coded by the Supplier/ Sub Supplier to avoid possibilities of mix-up. Details of color code shall be as specified in purchase documents. As per approved ITP - TPIA may witness and identify the same in the IRN/IC prior to the dispatch of these materials.

Piping materials shall be stamped / identified by the manufacturers as per the material specifications. This is to ensure availability of the markings (viz., heat number, material specification etc.) after welding with other materials at site during traceability verification. It is mandatory to check these markings and location, to be in line with Code & Material Specifications, by the TPIA before dispatch.

TPIA shall carry out hard / Low Stress stamping / Tagging / stenciling for Identification, Traceability and Verification. Same shall be mentioned in IRN/ IC of TPIA suitably.

Positive Material Identification (PMI) testing shall be carried out as per OWNER/PMC specifications / approved PMI procedure, and ITP/QCP by supplier at their works. TPIA shall witness PMI as per approved ITP/QCP.

Supplier & TPIA to ensure performance/ compliance/ completeness of inspection activities i.e. Review, Witness and Hold in sequential order by LTHE/ CONTRACTOR/PMC/ OWNER w.r.t. approved ITP before proceeding to next activity. Review of documents shall be carried out progressively as and when activities are scheduled and not at the end i.e. completion of all manufacturing activities

Any deviation or non-conformity with respect to the approved Documents is not acceptable. The same shall be reported by the Supplier / TPIA immediately to CONTRACTOR for resolution.

Supplier shall take special care to protect the final painting of the equipment / items during handling, storage, loading etc., in order to minimize touch up and re-work at site.

20 WITNESS INSPECTION AND TESTS

TPIA shall issue Field Inspection Report (FIR) / Inspection Report (IR) for each visit. The same shall submitted to LTHE and uploaded in ICMS immediately upon completion of inspections.

In case, inspection cannot be completed or undertaken due to reasons such as non-readiness of material, back up documents etc., such incidents and reasons shall be recorded in FIR / IR, and may be jointly signed by the Supplier and the TPIA.

TPIA / Supplier / Sub-supplier shall submit Inspection reports & back up inspection documents to CONTRACTOR within 1 calendar days of completion of inspection activity for review. Primary responsibility of submission of documents shall be of Supplier.

Before judging final acceptance of the equipment and materials at the time of final inspection, supplier's documents shall be reviewed by the inspector to verify the following:

- All specified inspections and testing have been completed.
- Results of inspections and testing conform to specified requirements.
- Documents have been prepared to prove that the finished product meets the specified requirements.

21 SUPPLIER NON-CONFORMANCE

Supplier / TPIA shall inform CONTRACTOR any non-conformity detected during Fabrication and/or detected by Inspector during his witness inspection within 24 hours of noticing.

If a non-conformance is found, Supplier shall immediately segregate the equipment or materials.

Based on the instructions of the CONTRACTOR's inspection engineer, visiting inspector shall ensure the rectification and immediately report the results to the CONTRACTOR's inspection engineer.

Supplier shall carry out the required/necessary actions for the rectification, modification or discard, etc., in accordance with the agreed disposition plan.

Supplier need to perform root cause analysis and revert within 72 hours of issuance of NCR for review & approval of originator and CONTRACTOR.

Project Quality Manager in consultation with Project Manager (if felt necessary) shall take remedial action to correct non-conformance. NCR need to be closed out by originating

organization. The complete management of Non-conformance shall be done as per flow chart provided in attached EXHIBIT-II.

22 INSPECTION CLEARANCE

CONTRACTOR/TPIA/OWNER/PMC reserves the right to verify the compilation of documentation during stage inspection and / or final inspection for all items. Supplier shall necessarily co-operate during such verification and comply with the specified requirements.

TPIA shall ensure and declare compliance to the approved ITP / QCP and make it part of the Inspection Certificate / Inspection Release Note. No material shall be shipped with open NCR.

Supplier/ Sub Supplier shall be responsible for obtaining EIL endorsement on IRN issued by TPIA for all category A & B Items / Equipments Supplier shall forward Inspection Release Note / Certificate issued by TPIA (PMC endorsed IRN for cat A & B) along with following reports as minimum to CONTRACTOR's discipline Inspection Engineer with copy marked to CONTRACTOR's Project Quality Assurance Manager.

- **Static item:** - IRN & IC of TPIA, Hydro-test reports, Final dimensional Reports, PWHT, Painting Reports, PMI reports, and MDR in case of Coded equipment.
- **Machinery Item:** - IRN & IC of TPIA, Hydro-test reports, Final Dimensional Reports, Painting Reports, PMI reports, Performance, Mechanical Run test, Unit test Reports.
- **E&I Item:** - IRN & IC of TPIA, Final Dimensional Reports, Painting Reports, PMI reports, Functional Test Reports, Statutory Certificates (As applicable).
- **Piping and bulk:** - IRN & IC of TPIA, Manufacturer TC's, Heat Treatment charts, Final dimension reports, PMI report etc.,
- **Piping Spool:** - IRN & IC of TPIA, Final Dimensional Reports/PWHT, Painting Reports, Hydro-test report, History Sheet, PMI report.
- **Fabricated Structure:** - IRN & IC of TPIA, Final Dimensional Reports, Painting Reports, History Sheet.

- **Packages:** - Applicable above items and / or IRN & IC of TPIA, Final Dimensional Reports / PWHT, Hydro-test reports, Painting Reports, Functional test report, Statutory Certificates and other test reports / certificates as applicable.

22.1 Note: Supplier shall also submit specific additional documents if requested by contractor for clearance

22.2 Without written clearance from CONTRACTOR, items shall not be dispatched to site.

23 TPIA DOCUMENTS

TPIA shall ensure that the Inspection Release Note / Certificate issued by TPIA shall mention the following details as a minimum:

- Project Name: VISAKH REFINERY MODERNIZATION PROJECT (VRMP)
- Package: EPCC 3 - RESIDUE UPGRADATION FACILITY (RUF)
- OWNER: Hindustan Petroleum Corporation Limited (HPCL).
- PMC: M/s ENGINEERS INDIA LIMITED.
- PMC Job No. : B016
- CONTRACTOR: L&T Hydrocarbon Engineering Limited (LTHE).
- CONTRACTOR's PO No. Including Amendments if any and Date
- Manufacturer's name
- Date & Place of Inspection
- Sub order No (supplier order reference for sub-ordered items)
- Approved PR / MR Specification No.(including Rev. No. and Date)
- Documentation approval status i.e. Approved Drawing No. (if applicable) including Approval, Rev. No. and Date
- Approved Inspection & Test Plan / QAP No.
- Item / Equipment description with TAG No (If applicable) as per PO.
- A list of MTC Nos for Bulk items
- Details of stages witnessed and documents reviewed as minimum as per approved ITP/QAP. i.e. Detail reference to Inspection Test Plan stage
- Written details of work performed and acceptability to acceptance criteria
- Intervention points that have been waived (if any)

- Quantity as ordered inspected and released.
- Status of the order whether complete or incomplete
- Problems/areas of concern
- Deviations, shortages if any.
- New and outstanding non-conformances
- Certification on IR / IRN's as **"INSPECTION HAS BEEN CARRIED OUT BASED ON CODE1/R DRAWING & APPROVED ITP."**
- *Inspector name with stamp and signature on inspection reports & IRN.*
- TPIA to issue separate release note (in detail) as required for mandatory and E & C spares indicating item description, Qty., etc.
- TPIA shall furnish Original hard copy and soft copy of IRN, IC.

24 FINAL DOCUMENTATION

Supplier shall submit index of final documentation to CONTRACTOR for review and approval. This index shall be in line with project approved format. This Index shall be approved by CONTRACTOR.

After Approval of index, Supplier shall submit soft copies of final documents as per approved INDEX for all stage and final inspection reports including approved procedures, material test certificates, test reports etc., for review and acceptance of CONTRACTOR as per approved ITP/QCP as minimum.

After acceptance of final documents (soft copy), Supplier shall submit number of hard copies of final documents as per the approved PR/MR & PO terms and conditions.

Suppliers shall submit the Inspection certificates issued by Statutory Inspection Authority as per the approved PR.

Suppliers shall compile separately the MTC's and IRN / Inspection Certificate for spares and submits along with the final documentation. The final documents supplied shall be duly endorsed by LTHE / TPIA as "DOCUMENTS VERIFIED AND CERTIFIED FOR COMPLETENESS AS PER CONTRACT."

25 EXHIBIT I – TYPICAL INSPECTION AND TEST PLAN FORMAT

STANDARD INSPECTION AND TEST PLAN FOR *(Mention Product Name)*

Purchaser :		Item Description	
Project :		ITP Document No.	Rev. 01
Client :		MR/PR/Specification No.	
L&T Job No. :		Inspection Category	
Vendor :		Name of TPIA	
Vendor Job No. :		ITP approval by	L&T & Client

Sl. No.	Inspection Activity	Extent of Check	Reference Document	Acceptance Criteria	Form to record results of compliance	Inspection Agency				Remarks
						Sub-Supplier	Vendor	L&T/ TPIA	Clelnt	

Legends:: H - Hold ; **W** - Witness ; **R** - Review ; **M** - Monitor , **IC** - Inspection Certificate, **IRN**- Inspection Release Note

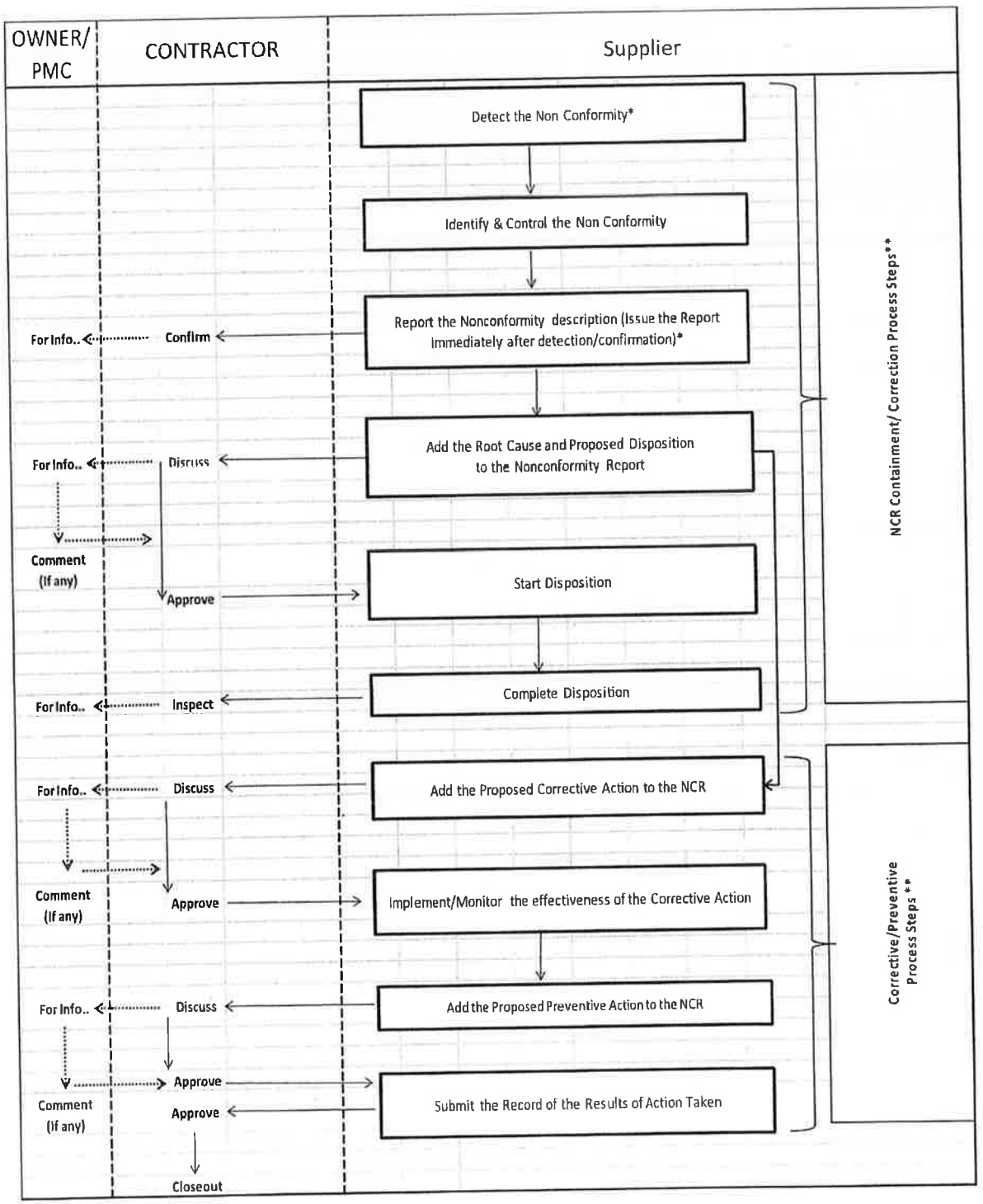
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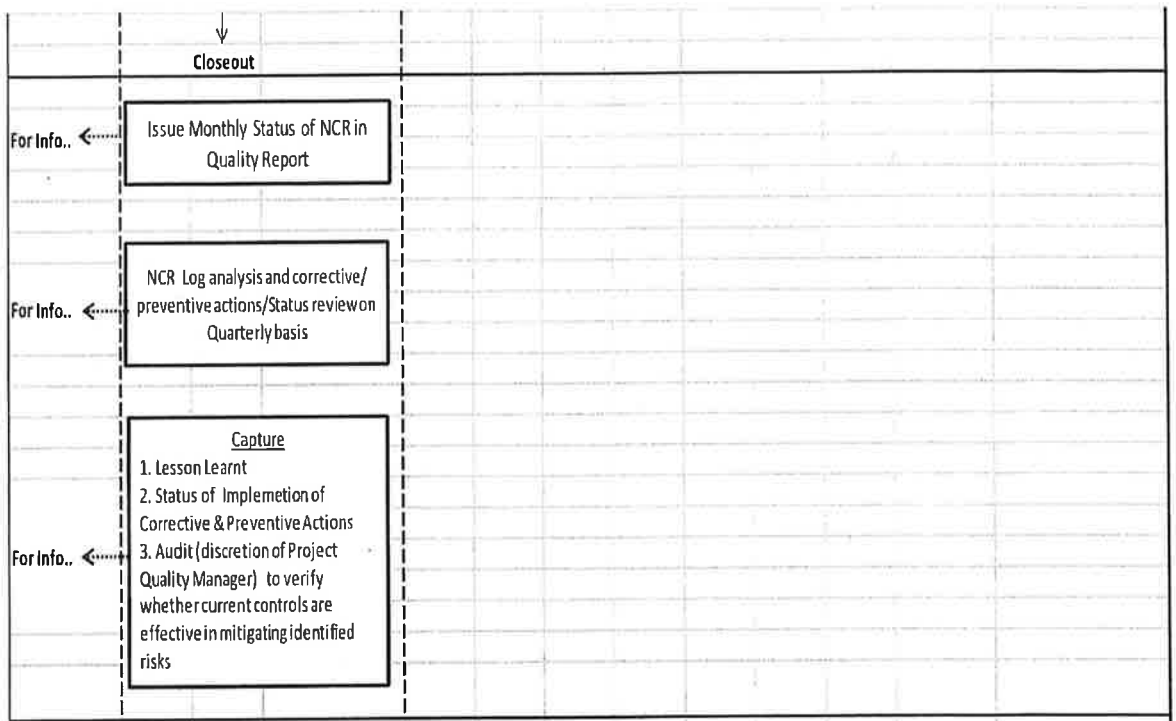
ITP Prepared by (Vendor)	ITP Approved by (L&T)	ITP Approved by (Client)

Inspection and testing activities completed

Signed on behalf of:		Sub-Supplier	Supplier	3rd Party	L&T	Company	Other
Signature:							
Print Name:							
Date:							

26 EXHIBIT II – FLOW CHART FOR NCR MANAGEMENT





a)*Detection of Non Conformity & Reporting of Non conformity shall be done either by Supplier or CONTRACTOR appointed Inspection agency or OWNER/PMC appointed inspection Agency or OWNER/PMC

b) **NCR Catagorization:

1. High Risk - Follow NCR Containment/Correction Process Steps & then Corrective/Preventive Process Steps.
2. Medium Risk - Follow NCR Containment/Correction Process Steps. Corrective/Preventive Process Steps under discretion of Project Quality Manager
3. Low Risk - Follow NCR Containment/Correction Process Steps. Corrective/Preventive Process Steps under discretion of Project Quality Manager

c) ** For Medium or Low risk NCR, Project Quality Manager may decide Corrective/Preventive action based on Repetative Items being ordered on the same Vendor/ reoccurance of similar NCR , thereby eleminating chances of Potential Non conformity